

Extending NLP heuristics to Speaker recognition using Discrete Speech Units

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M1 Artificial Intelligence

Research Objectives

- Classify speakers using DSUs.
- Determine what information is extracted using DSUs: text features, or also phonetic ones.
- Suggest a model that allows for quick and effective classification of speakers in comparison with known methods.

Overview

Models, baselines, datasets

Results

Discussion

1. Models and Baselines

Baseline

Continuous features

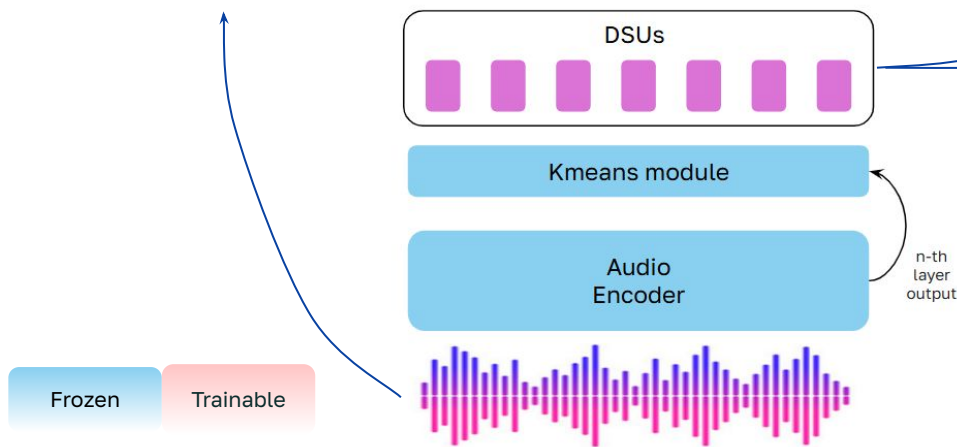
Mel spectrogram + RNN

Models

Discrete Speech Units

CV, TF-IDF +
classification

Embeddings + RNN



For decisions regarding DSU extraction, we follow:
Pooneh Mousavi, Jarod Duret, Salah Zaiem, Luca Della Libera, Artem Ploujnikov, Cem Subakan, & Mirco Ravanelli. (2024). *How Should We Extract Discrete Audio Tokens from Self-Supervised Models?*.

"The quick brown fox jumps over the lazy dog"

2. Datasets

Steps followed for data gathering (LibriSpeech*):

1. Concatenated all the version (100h / 360h / 500h)
2. Selected 30 speakers which have the most number of audios files associated to them.
3. Split the entire data into train-dev-test split of 88%-6%-6% respectively sampled equally across speakers.

Steps followed for data gathering (VCTK**):

1. Select instances where the transcript is repeating at least 4 times.
2. Select the top 30 speakers, frequency wise.
3. Split the entire data into train-dev-test split of 88%-6%-6% respectively sampled equally across speakers.

*Pratap, V., Xu, Q., Sriram, A., Synnaeve, G., & Collobert, R. (2020). MLS: A Large-Scale Multilingual Dataset for Speech Research. In *Interspeech 2020*. ISCA.

**Veaux, C., Yamagishi, J., & King, S. (2013). The voice bank corpus: Design, collection and data analysis of a large regional accent speech database. In *2013 International Conference Oriental COCOSDA held jointly with 2013 Conference on Asian Spoken Language Research and Evaluation (O-COCOSDA/CASLRE)* (pp. 1-4).

2. Datasets

- **LibriSpeech**

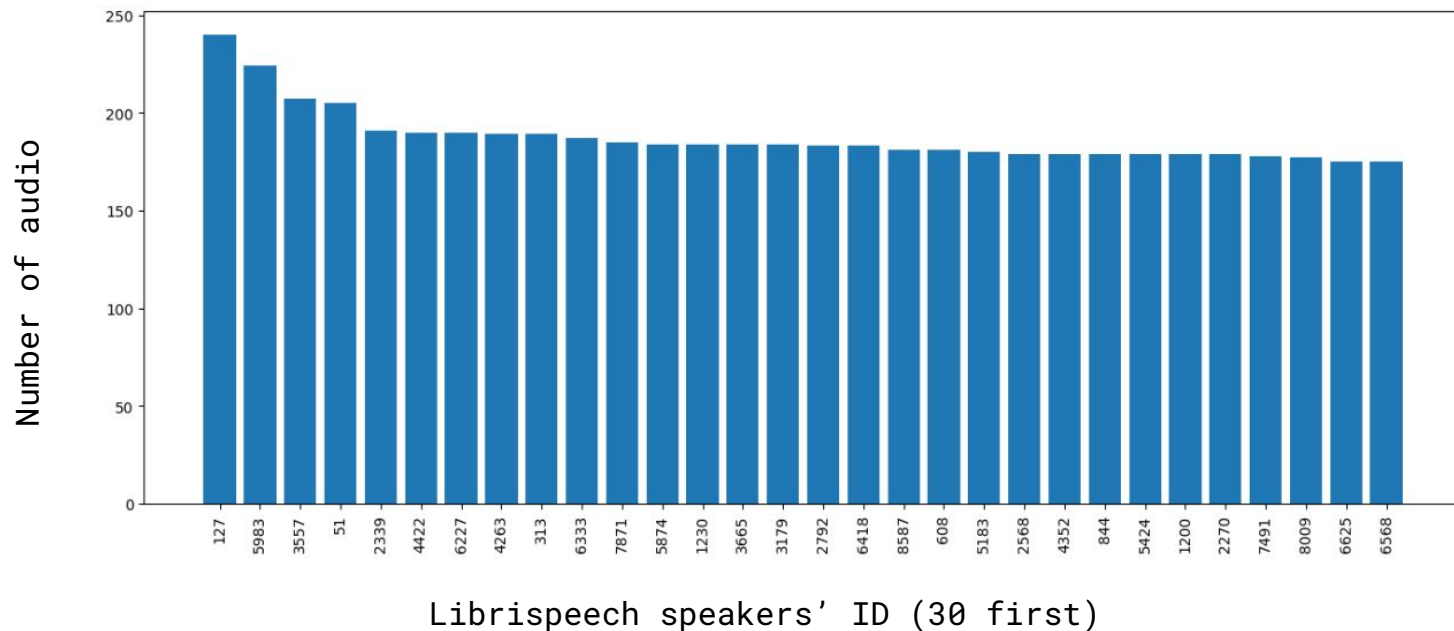
- Train set: 4933 rows
- Dev set: 335 rows
- Test set: 352 rows

- **VCTK**

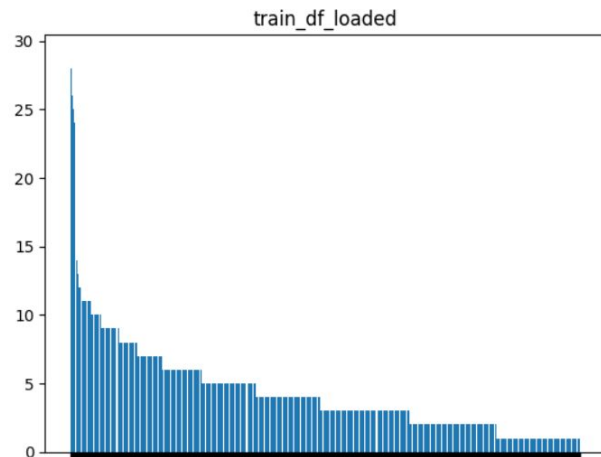
- Train set: 10050 rows
- Dev set: 558 rows
- Test set: 571 rows

- **file**: the path to the downloaded audio file
- **text**: the transcription of the audio file
- **speaker_id**: unique id of the speaker
- **DSU**: compact representations for speech signals that were obtained by k-means clustering of encoded audio
- **DSU_Deduplicated**: DSUs representation after data preprocessing

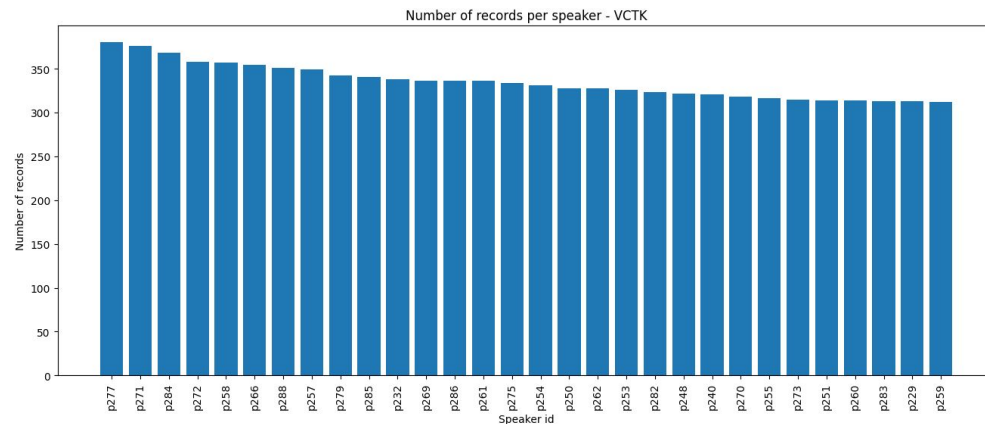
2. Dataset: LibriSpeech



2. Dataset: VCTK

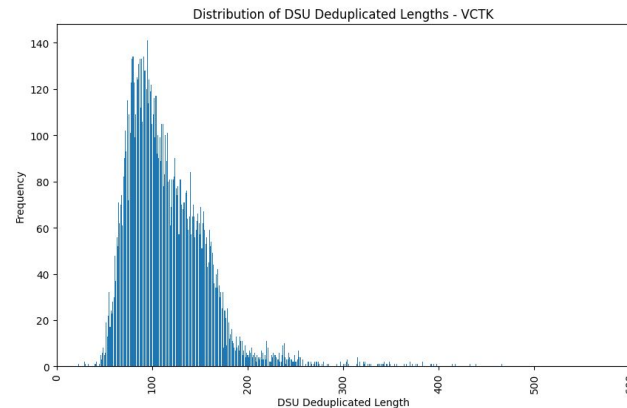
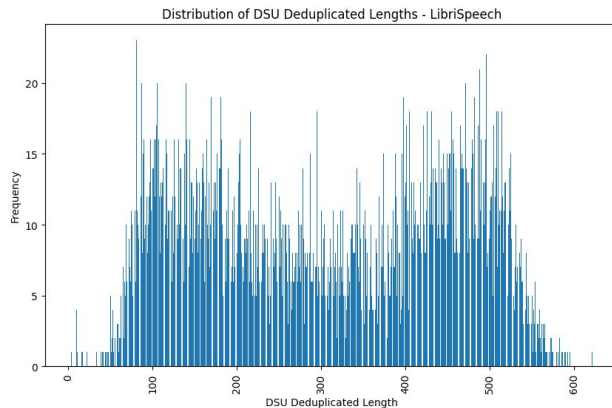
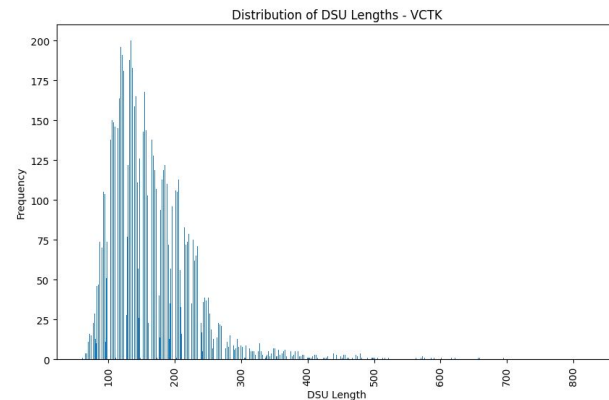
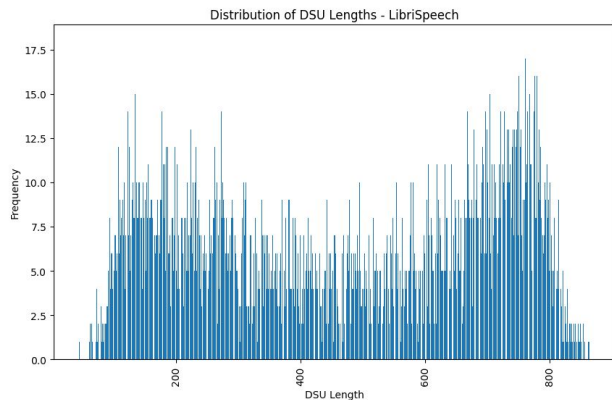


Train data transcript repetition in VCTK

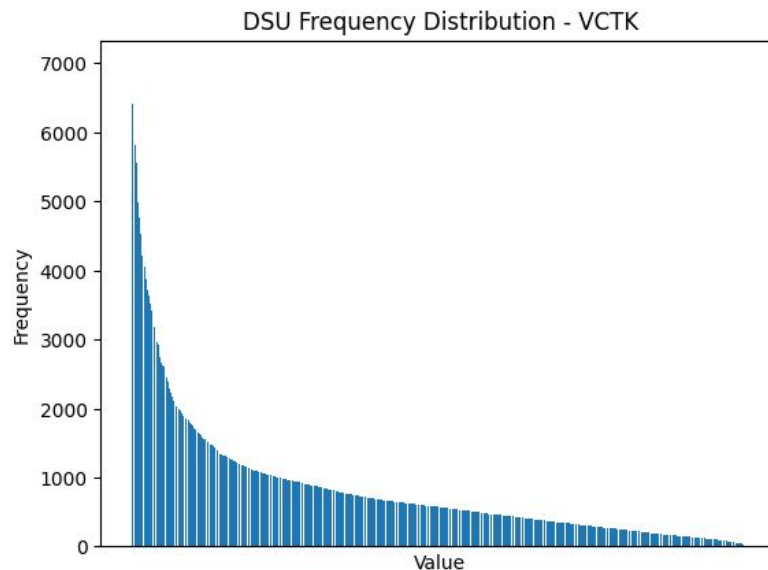
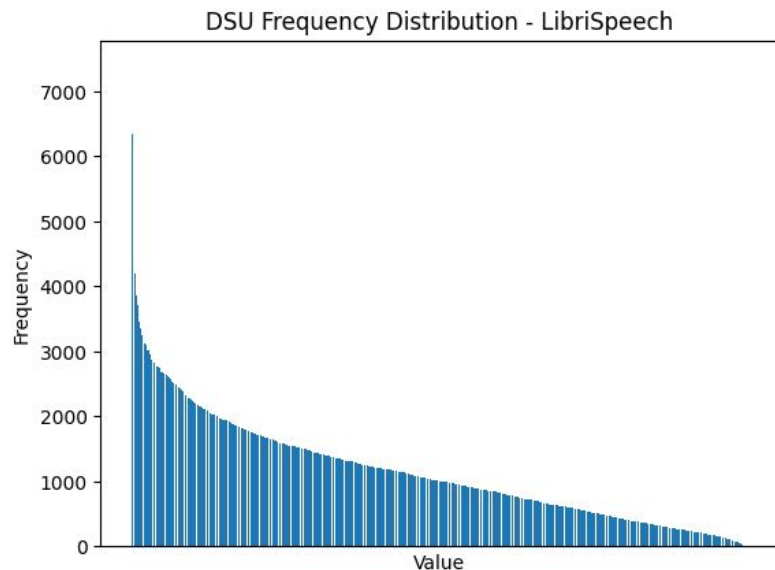


Audio frequency per speaker in VCTK (train set)

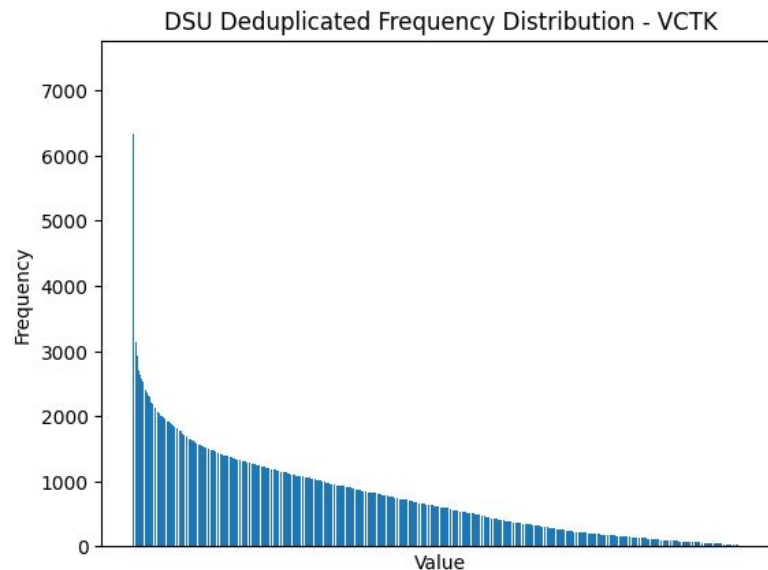
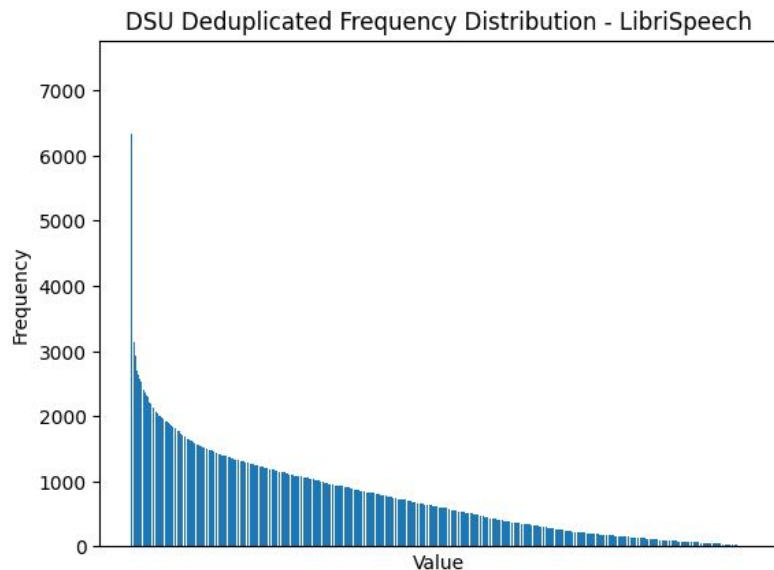
2. Dataset: Length of DSUs



2. DSU Frequency Distribution



2. DSU Deduplicated Frequency Distribution

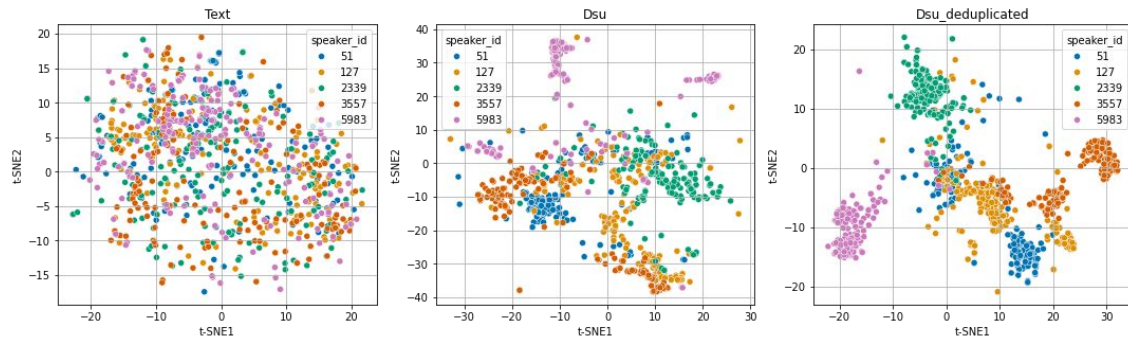


2. Results overview (Vectorizers)

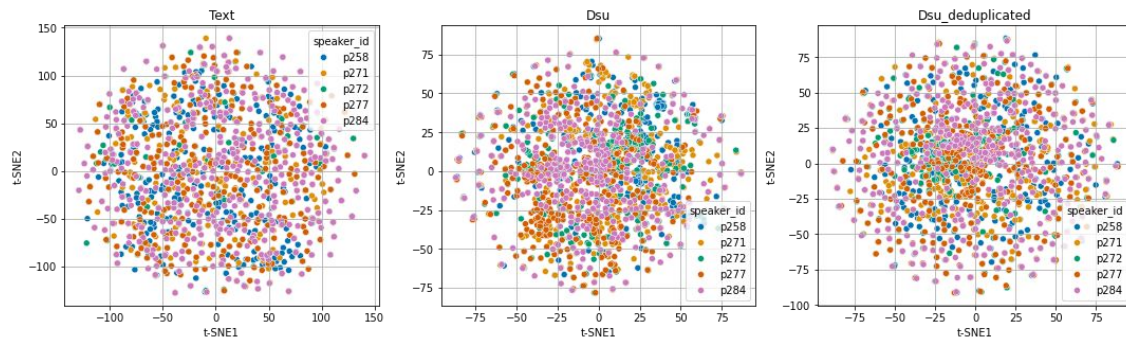
Features	F1-score			
	CounterVecorizer		TF-IDF	
	MNB, unigrams	MNB, bigrams	MNB, unigrams	MNB, bigrams
LibriSpeech				
DSUs	0.9624	0.9800	0.9575	0.9800
DSUs Deduplicated	0.9681	0.9828	0.9594	0.9772
Transcript (text)	0.5431	0.4090	0.5615	0.4480
Continuous	–	–	–	–
VCTK				
DSUs	0.6125	0.6706	0.5667	0.6599
DSUs Deduplicated	0.6679	0.7071	0.6352	0.6934
Continuous	–	–	–	–

2. Results: CountVectorizer, t-SNE

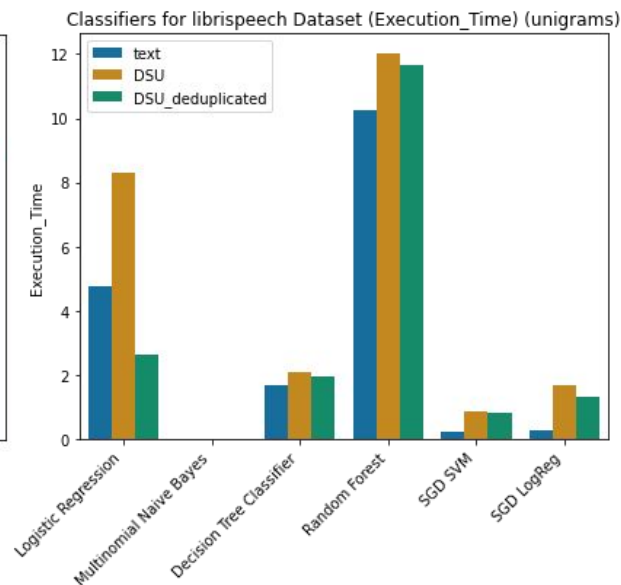
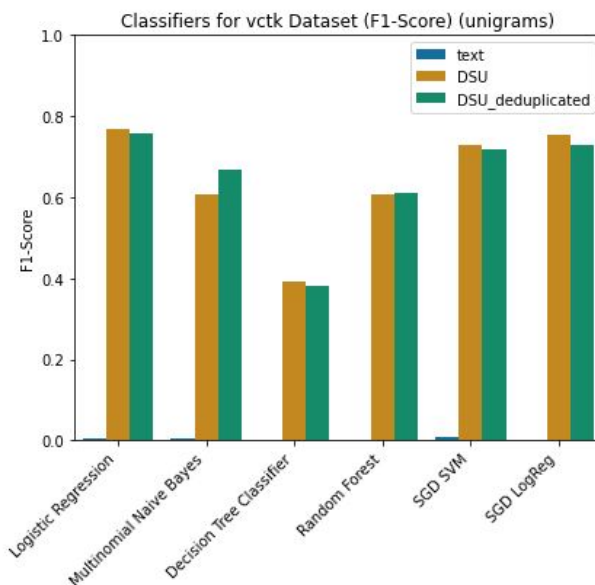
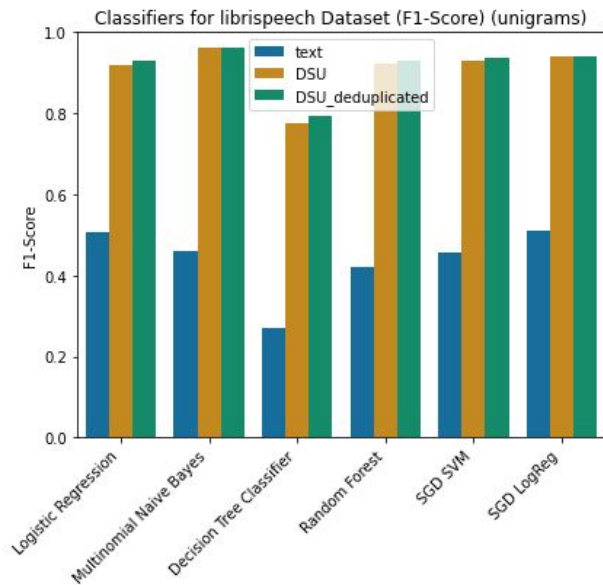
t-SNE Visualization - Librispeech (n_grams = (1, 1))



t-SNE Visualization - Vctk (n_grams = (1, 1))



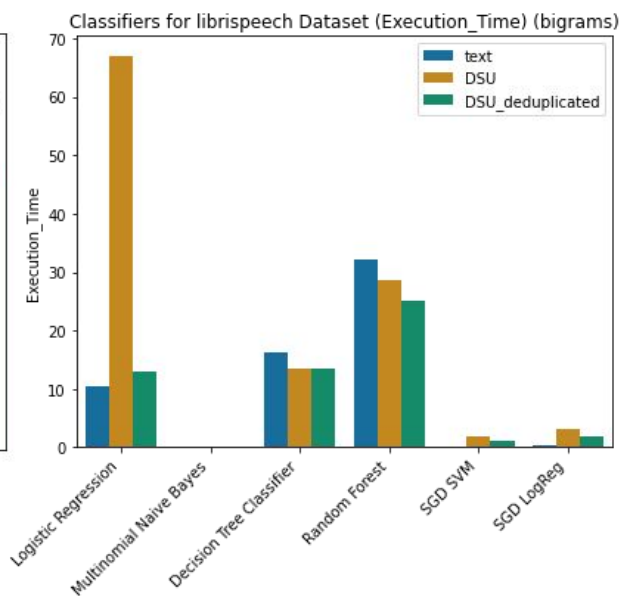
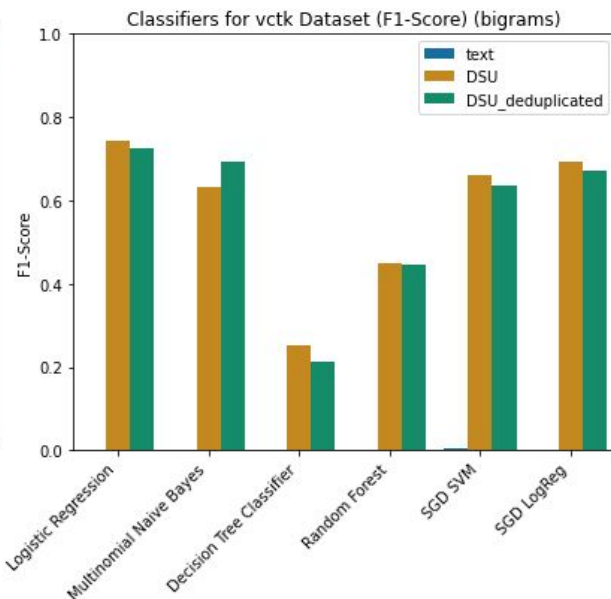
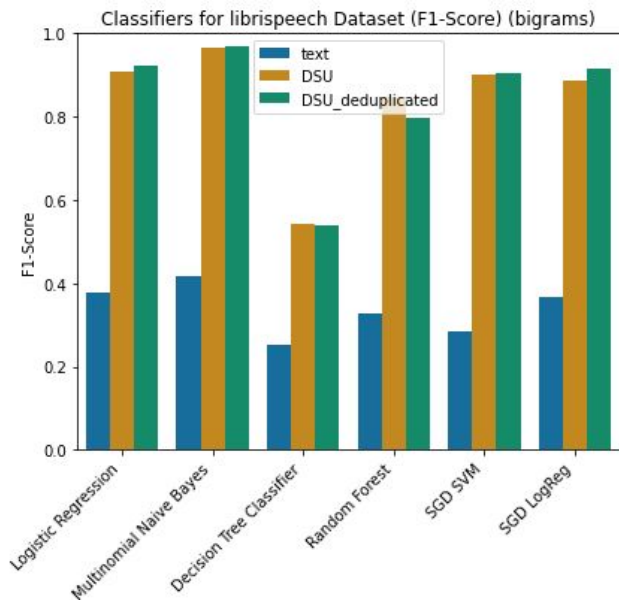
2. Results: CountVectorizer, unigrams



F1-score

Execution Time (s)

2. Results: CountVectorizer, bigrams

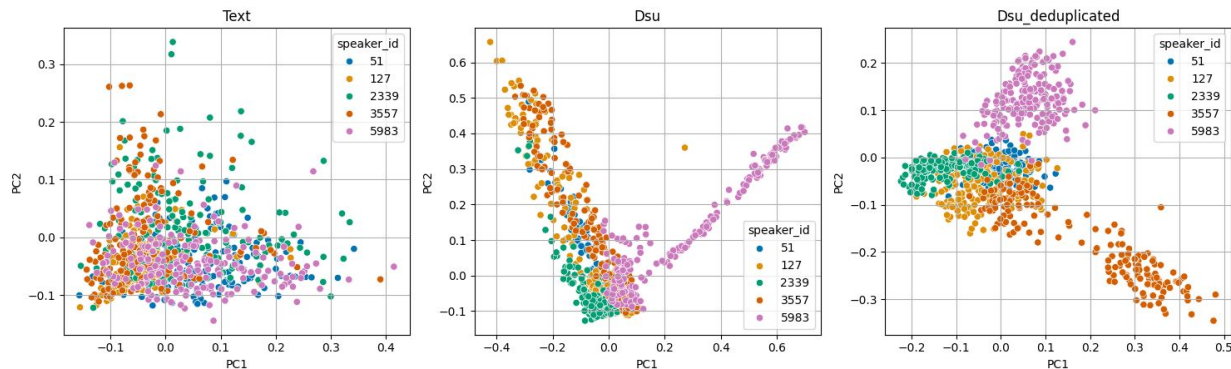


F1-score

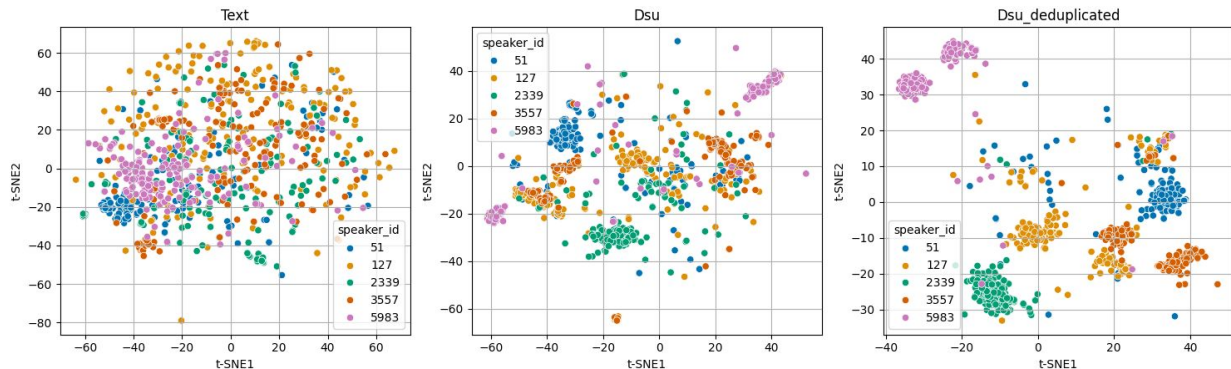
Execution Time (s)

2. Librispeech: TF-IDF, unigrams

PCA Visualization - Librispeech (n_grams = (1, 1))

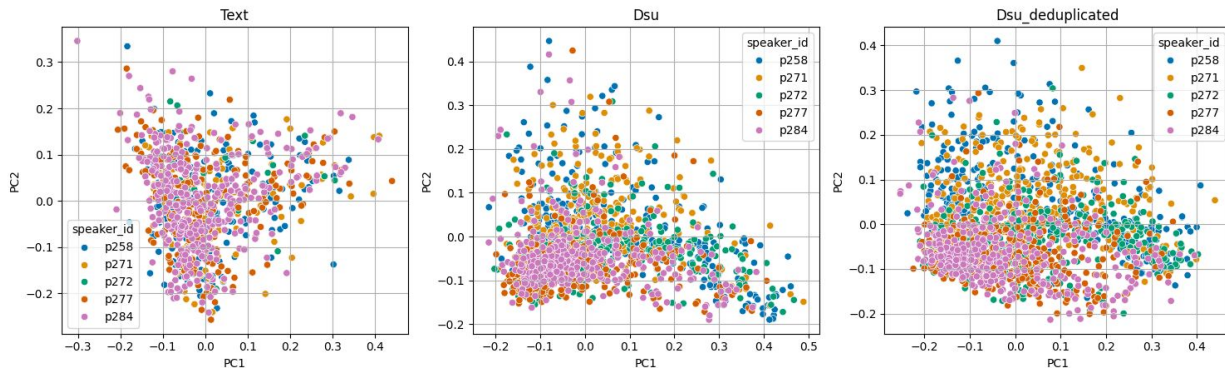


t-SNE Visualization - Librispeech (n_grams = (1, 1))

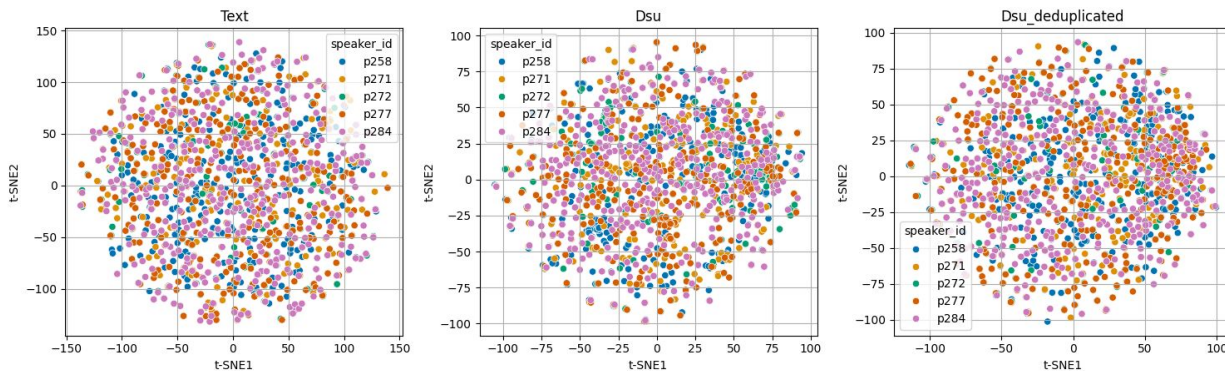


2. VCTK: TF-IDF, unigrams

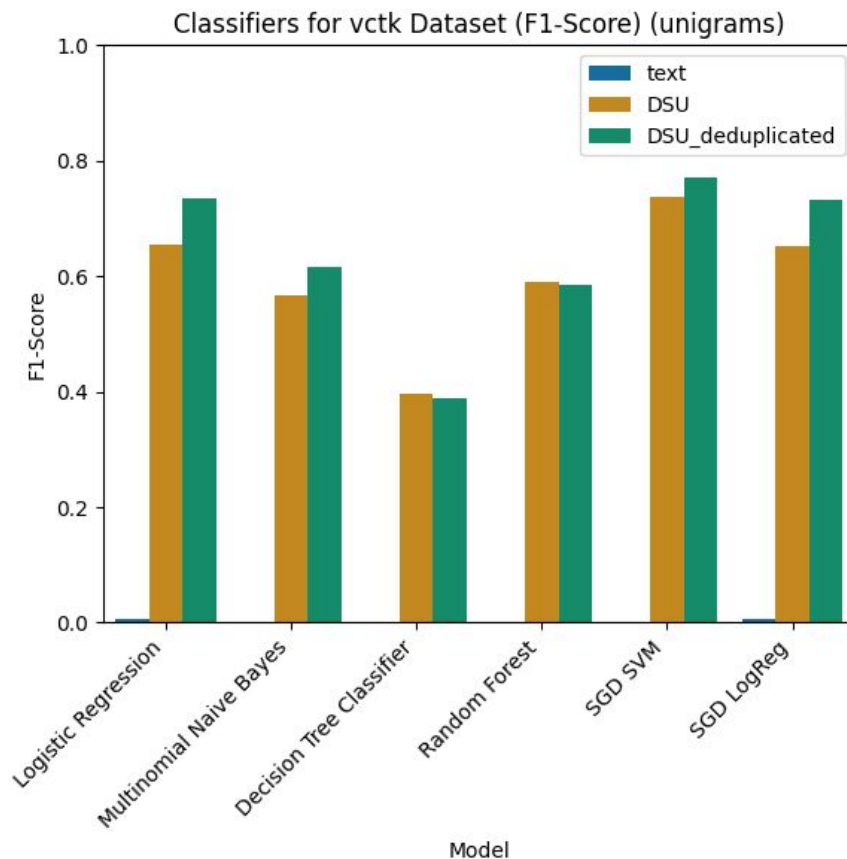
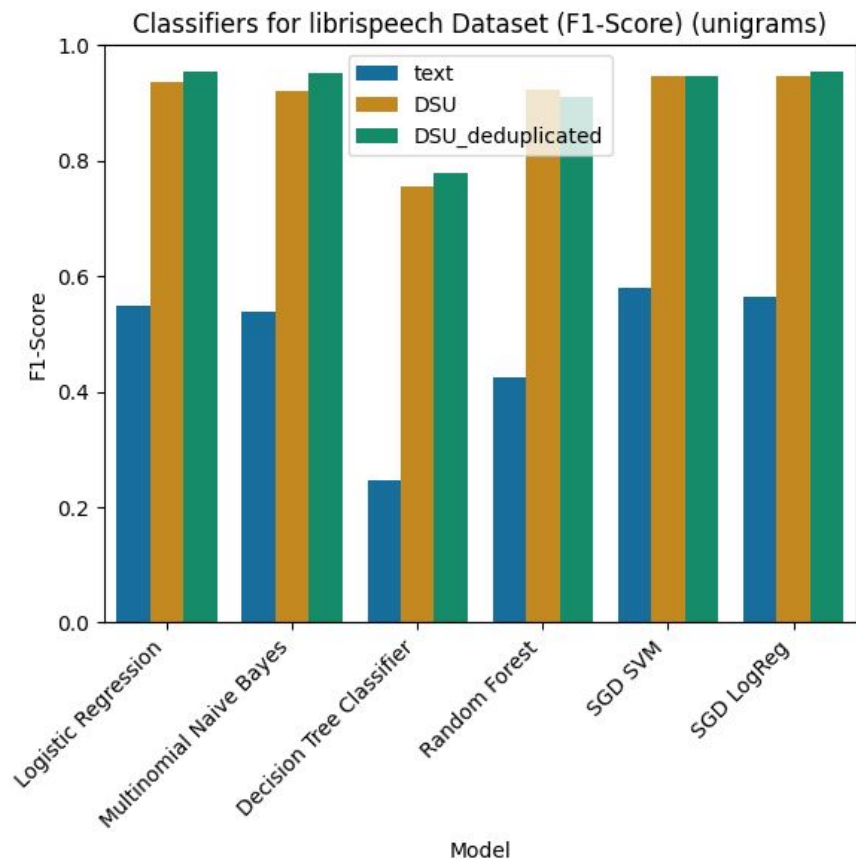
PCA Visualization - Vctk (n_grams = (1, 1))



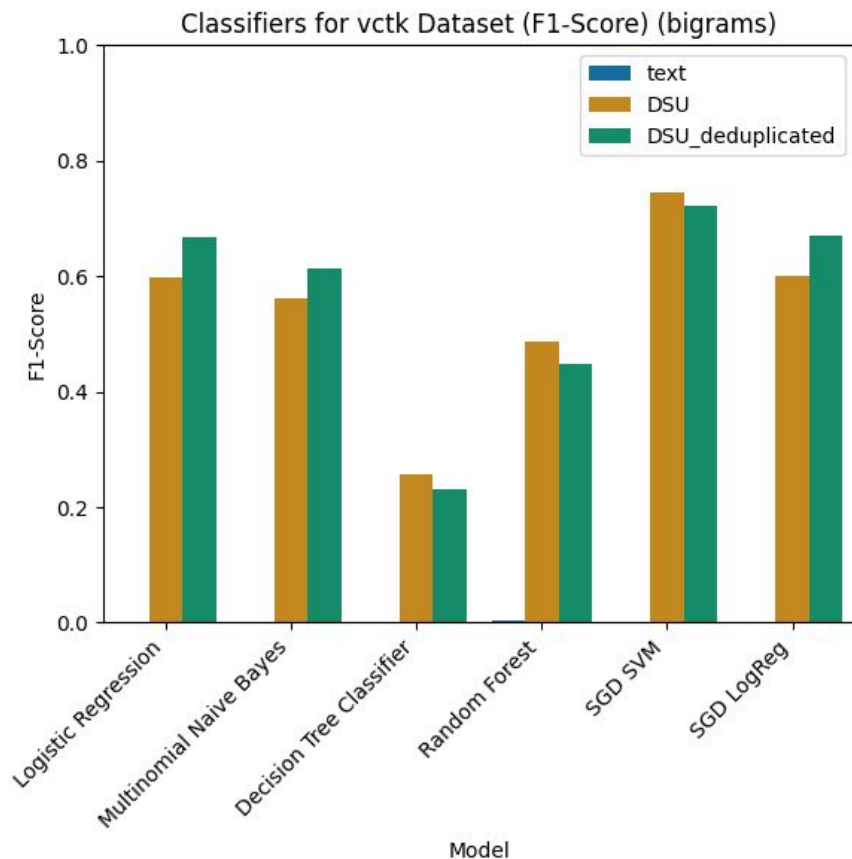
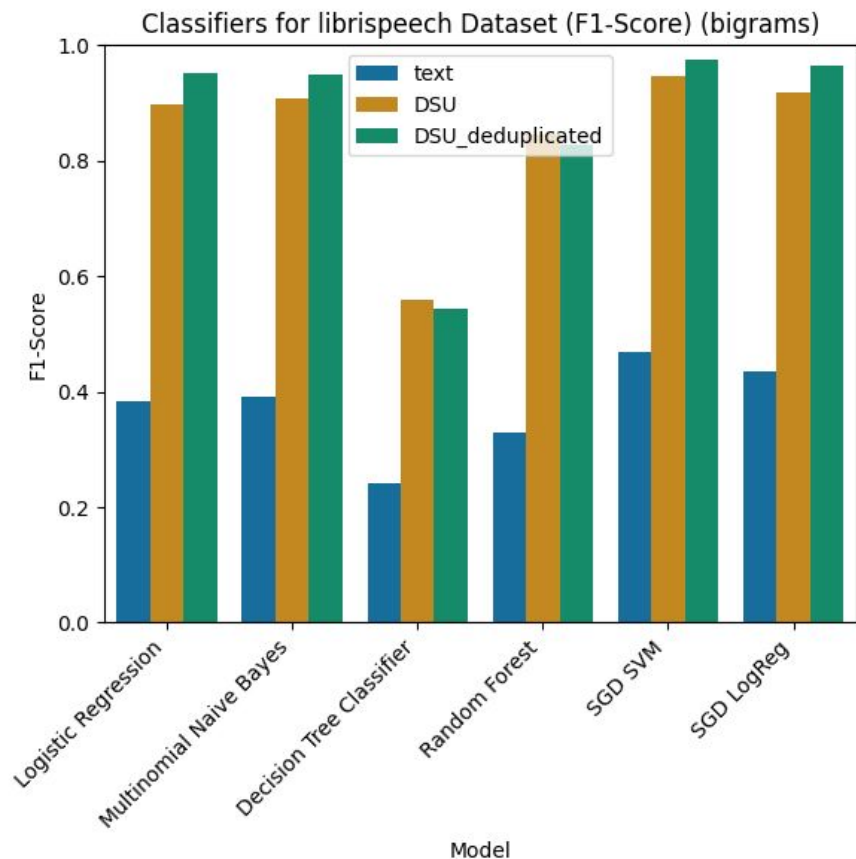
t-SNE Visualization - Vctk (n_grams = (1, 1))



2. Results: TF-IDF, unigrams



2. Results: TF-IDF, bigrams

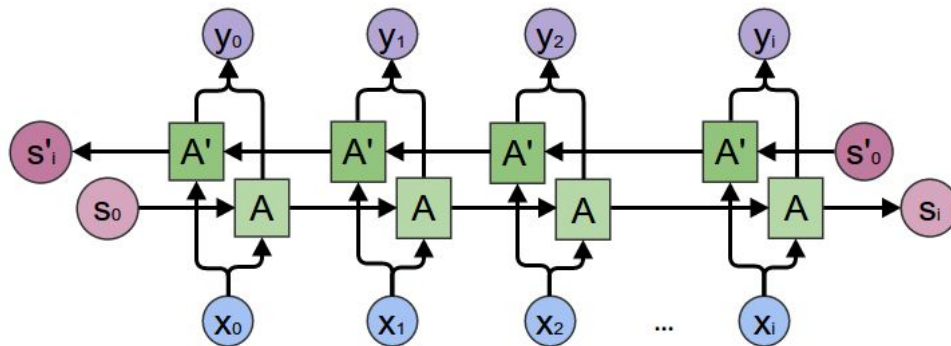


2. Results: RNN on continuous features

```
SpeakerRNN(  
  (rnn): RNN(1024, 256, num_layers=3, batch_first=True, bidirectional=True)  
  (classifier_layer): Linear(in_features=512, out_features=30, bias=True)  
)
```

```
SpeakerLSTM(  
  (embed): Embedding(2001, 128, padding_idx=2000)  
  (lstm): LSTM(128, 128, num_layers=4, batch_first=True, dropout=0.2, bidirectional=True)  
  (dropout): Dropout(p=0.2, inplace=False)  
  (classifier_layer): Sequential(  
    (0): Linear(in_features=256, out_features=128, bias=True)  
    (1): Tanh()  
    (2): Linear(in_features=128, out_features=64, bias=True)  
    (3): Tanh()  
    (4): Linear(in_features=64, out_features=32, bias=True)  
    (5): Tanh()  
    (6): Linear(in_features=32, out_features=30, bias=True)  
  )  
)
```

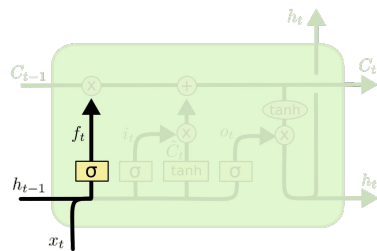
Sidequest: RNN Architecture



$$\boxed{h_t} = \boxed{f_W}(\boxed{h_{t-1}}, \boxed{x_t})$$

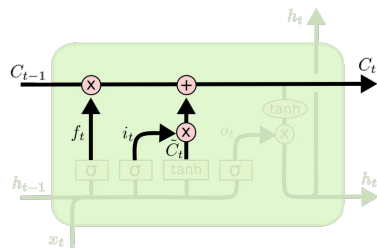
new state some function with parameters W old state input vector at some time step

Sidequest: LSTM Architecture



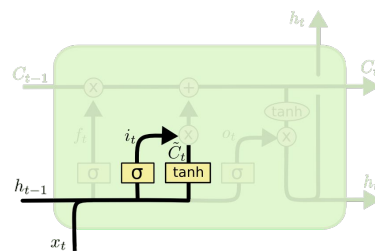
forget gate layer

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$



cell state update

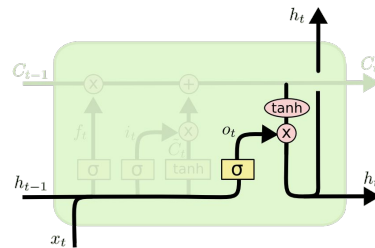
$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$



input gate layer

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

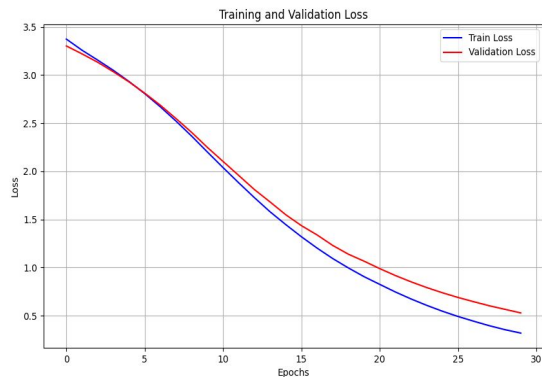


Hidden state update

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t * \tanh(C_t)$$

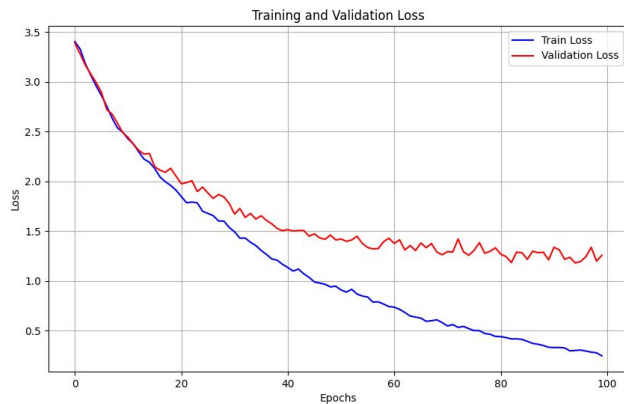
2. Results: RNN on continuous features



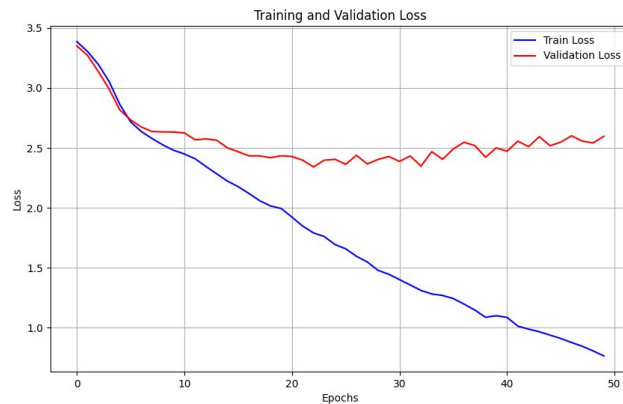
RNN on Continuous features (VCTK)

Dataset	Accuracy	Precision	Recall	F1 Score
VCTK	0.8932	0.9021	0.8932	0.8939
LS	0.9403	0.9455	0.9403	0.9403

2. Results: LSTM



LSTM on LibriSpeech



LSTM on VCTK

Dataset	Accuracy	Precision	Recall	F1 Score
VCTK	36.60	38.13	36.60	36.77
LS	73.58	75.22	73.58	73.48

2. Results: Overall

Features	F1-score				
	CounterVecorizer		TF-IDF		RNN
	MNB, unigrams	MNB, bigrams	MNB, unigrams	MNB, bigrams	
LibriSpeech					
DSUs	0.9624	0.9800	0.9575	0.9800	–
DSUs Deduplicated	0.9681	0.9828	0.9594	0.9772	0.7348
Transcript (text)	0.5431	0.4090	0.5615	0.4480	–
Continuous	–	–	–	–	0.9403
VCTK					
DSUs	0.6125	0.6706	0.5667	0.6599	–
DSUs Deduplicated	0.6679	0.7071	0.6352	0.6934	0.3677
Continuous	–	–	–	–	0.8939

Discussion

1. Text features are not enough, phonetics are important
2. Continuous features outperform DSU-based methods
3. DSUs are prone to overfitting in limited data regime
4. TF-IDF and CV provide a better, much simpler alternative in this setting.

Conclusion

1. DSUs for Speaker Classification

- a. Effective in Librispeech but struggled in VCTK due to limited data and sentence diversity.
- b. Overfitting mitigated with LSTM models, but generalization remained a challenge.

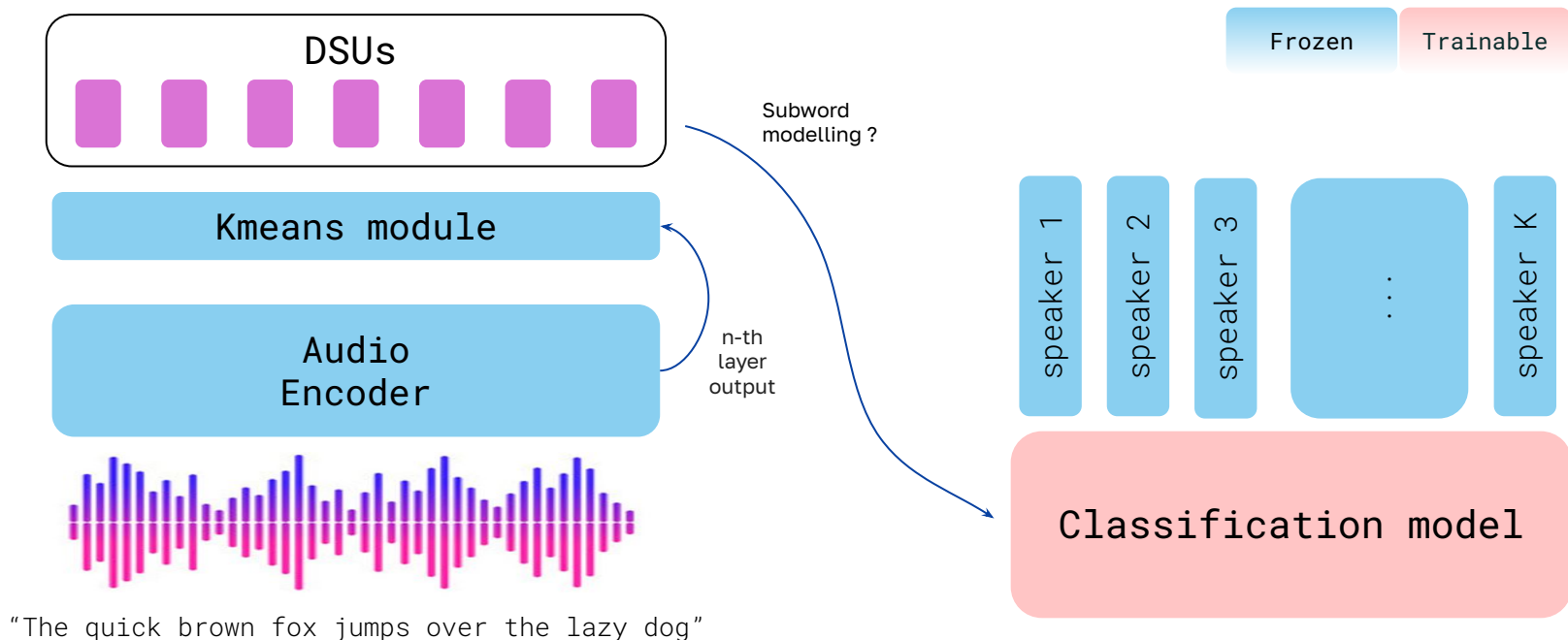
2. Comparison with Text & Continuous Features

- a. Text-based features alone were insufficient for speaker identification.
- b. Continuous speech features outperformed DSUs, emphasizing the role of phonetic variations.

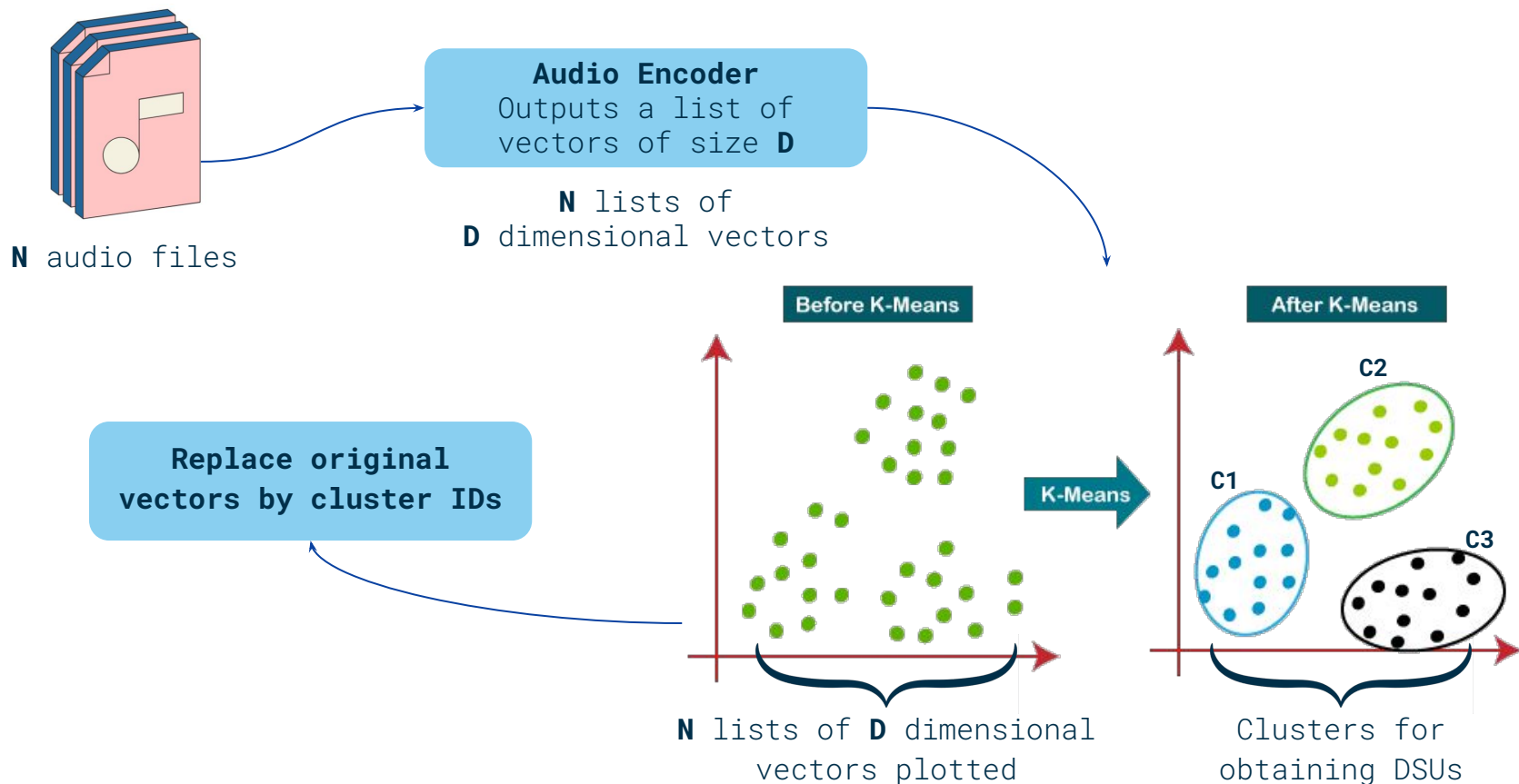
Thank you!

1. Problem statement: technique

Classify the speakers (speakers ID) using Discrete Speech Units (process them like a text)



1. Problem statement: technique



1. Problem statement: DSU example

```
3409 917 2670 1784 559 1579 1371 4426 365 356 3626 4433 462 2940 2186 551 3677 487 2863 4911 3805 3939
1434 2573 701 1567 95 432 3049 3326 2212 2389 4793 4706 1878 1686 2415 2017 1036 2540 4101 4202 2183
114 63 665 3419 564 2222 3607 3423 457 1237 2844 963 568 1907 223 4705 3050 80 3441 2552 4228 2092 3174
4172 132 3846 1593 436 1134 939 615 2057 516 1695 741 2872 910 4446 1001 3374 9 1084 3271 742 4092 1643
1690 911 1086 518 445 538 4505 3773 1011 4100 162 2324 1365 245 1 781 4112 591 4694 2838 300 4895 114
63 698 2655 437 3350 4141 177 4730 2096 2812 231 1724 1644 1154 211 1532 4093 595 2559 1916 371 36 1187
1832 4637 1377 186 3111 260 2622 3374 4876 1069 313 620 2757 1350 2663 1743 4186 2313 1008 95 1723 1109
3374 4077 3702 1069 313 1892 1608 3393 3413 4269 2217 3279 3290 3535 303 1907 694 3991 3013 3040 2593
399 3295 399 4389 1286 2778 2827 4195 742 1910 2391 14 1097 4381 1461 604 1634 2660 546 432 563 1815
2832 1615 2158 3953 2591 2155 584 4251 1244 303 910 1170 4052 275 2420 240 1236 2122 3680 1271 1147
1581 2500 15 3071 374 742 374 2699 1016 87 3164 803 453 72 453 2154 527 3958 1586 1162 1389 125 577
1862 2047 4141 492 4354 3476 3548 1517 1037 1554 4433 2905 4707 2813 3839 339 4744 2959 4485 1829 3189
559 340 1579 3387 1371 2935 2735 365 1686 1145 4590 52 3118 2393 2561 4033 3514 1182 1697 4914 3086
2605 3771 4591 3189 1724 102 1082 2758 3474 3089 2489 3234 2027 273 2905 4260 2127 4131 1113 2251 4913
275 2420 3160 2883 616 2060 3646 1722 303 910 3683 1478 2812 3326 102 654 1215 2132 982 2283 3698 3038
892 2876 978 2292 1216 706 2186 4973 56 405 3009 3646 358 3079 752 3921 1464 719 2477 3350 4035 3102
177 4354 1449 2603 1330 4988 713 1107 143 275 486 1179 1573 3430 225 4780 3405 419 1840 222 636 1055
522 2527 4414 2878 171 4507 699 2151 4385 1761 3737 3945 3189 687 1908 1257 3336 1339 114 4738 2196
3188 3171 2168 3364 662 42 4035 1445 790 4403 2144 1004 2262 3648 1853 583 4907 2812 55 439 784 3392
916 4465 3687 2298 2659 2099 581 687 231 340 1579 1371 2935 2735 365 290 445 742 538 4505 3773 2898
3294 621 1311 634 4811 3557 399 3796 3587 1245 3913 532 828 2030 710 2725 2026 3958 4122 2144 1004 2262
1853 583 2096 742 1724 439 784
```